

Teach Me Where to Look: Dual-task Attention Training in Augmented Reality

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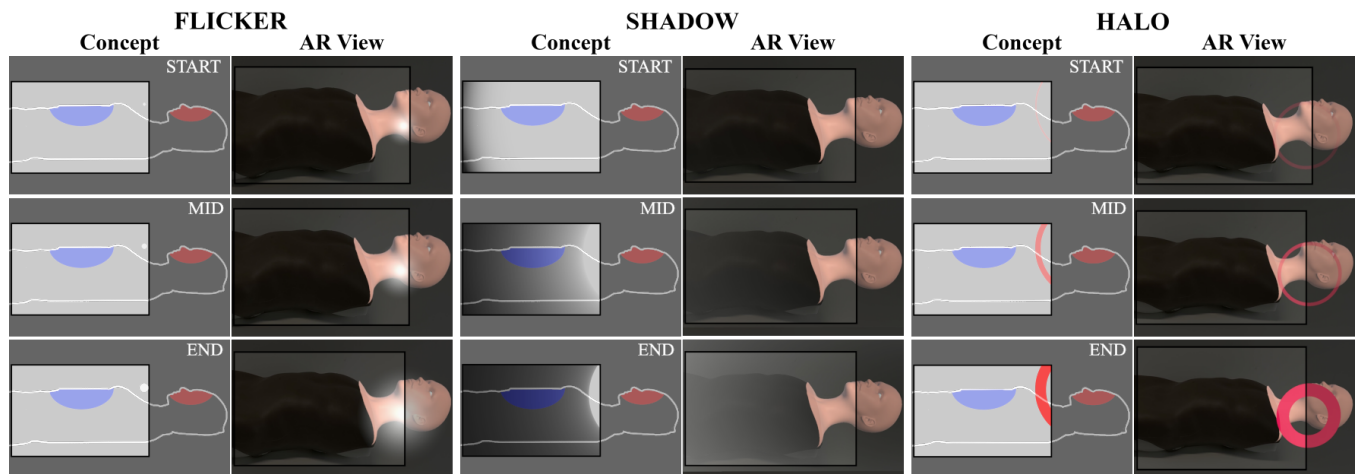


Figure 1: Visual attention guidance concepts. All cues are shown in concept and AR view (columns), and at three intensification stages (rows). The bright, black outlined area marks the HoloLens field of view, with the dark area for context. For the concepts, the working (blue) and target area (red) are marked. Visibility of all concepts is low in early stages (first row) and is increased for 20s (bottom row). Visual differences between the concept and AR result from the display parameters of the HoloLens.

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Abstract

Regular eye contact is essential in medicine to recognize signs of pain. However, it is difficult to remember this during training as attention is tied up in learning. While augmented reality (AR) has shown promising results for medical education, there is no training for attention allocation yet. Therefore, three auditory and three visual attention guidance tools in AR are evaluated for their use in medical dual-task training settings. In expert reviews with six participants in human-computer interaction and medical didactics,

advantages, disadvantages, and refinements for the cues were developed. For visual cues, an overt but less occluding cue was preferred for constant visibility of the primary task. A more diegetic cue design was proposed for the auditory cues to use a patient simulation as a reminder of the regular face glance. In general, several cues were found to be suitable for gaze guidance training, requiring only minor changes for improvement.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Interaction techniques**; • **Applied computing** → **Interactive learning environments**.

Keywords

Attention Guidance, Augmented Reality, Attention Allocation Training

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1 Introduction

The human face is an important medium for non-verbal communication. Difficulties in regularly looking at another person's face can be both permanent, e.g. due to developmental disabilities [27] and task-dependent, e.g. while focusing on something else [55]. In medical examinations, paying attention to the face is important so as not to overlook any signs of pain or discomfort. Based on McIsaac et al. [32], this can be defined as a dual-task scenario: Observation of the face with the goal of obtaining non-verbally communicated information, and the physical examination to recognize physical symptoms. While medical students learn the overall task of examining the patient, they find it difficult to perform these sub-tasks in parallel [43]. Related work has shown both the positive effects of augmented reality (AR) in medical training [33] and on attention guidance in dual-task scenarios [21]. However, less is known about how attention allocation to a specific region can be practiced and integrated into AR training situations. For this reason, we have compiled a series of well-known attention guidance cues. All tools were adapted for our dual-task setting with a haptic-cognitive primary and a visual-cognitive secondary task. In a first step towards the development of an integrable training tool, we evaluated the cues using an abstracted scenario in expert reviews. The individual suitability and necessary adaptations for subsequent quantitative studies were determined. With these findings, this work contributes to the general design of future AR applications using attention allocation. Here, not only medical applications are of interest, but also other multi-activity scenarios like learning to drive [20].

2 Related Work

This paper follows the taxonomy for attention-guiding methods of Rothe et al. [39]. Due to the context, visual and auditory cues of different awareness/directness levels will be presented in particular.

For visual cues, subtle and overt aspects are examined in more detail [35]. Furthermore, spatial auditory guidance and diegetic cues are presented.

Subtle visual guidance uses not consciously perceived cues, for example eye movement-dependent presentation [4], unnoticeable modulations [50] or by using perceptual properties [51]. Subtle guidance is less interrupting than overt cues [36], but performance can be worse in crowded environments [22]. Generally, there is a trade-off between guidance efficiency and disruptiveness of a cue [11]. Required intensities for effective guidance might reduce user enjoyment and comfort even for subtle cues [38]. Gaze-dependent color and luminance modulations can be presented as an animated peripheral stimulus to achieve a flickering effect, providing reliable gaze direction [4, 52]. Based on the higher temporal resolution of the peripheral field of view, cues can be perceived as flickering when in the periphery, but static when in the central field of view [51]. Other subtle cues use contrast modulation to create a blur effect in the context area of the focus object. Guidance was found to be effective but not always reliable, with users not being aware of the blur [50].

Subtle gaze guidance works best for in-view targets [29], however, movement towards the target can be used for out-of-view scenarios [15]. In general, subtle image changes are difficult to realize in optical see-through AR due to limited reality manipulation [46].

Overt visual guidance cues typically indicate the direction and position of the target indirectly, for example via an arrow [39]. However, more prominent versions of subtle cues can also be used, e.g. blurring [2] or fading to black [11]. The guidance concept Halo draws a circle around the target that extends to the edge of the screen [8]. This allows for a precise estimation of the target position [10]. Furthermore, the visualization can be easily modified, e.g. by adding animations [54] or changing display parameters [34]. Adaptions for 3D displays were made by using sphere projections [16, 17]. However, Halo leads to visual clutter, especially with multiple targets. To this end, the Wedge visualization was proposed, displaying isosceles triangles with one tip at the target position and two corners at the edge of the screen [19]. Both Halo and Wedge outperformed arrow-based cues, with Halo yielding lower directional error in target localization and Wedge performing better in terms of usability [17]. In virtual reality (VR), no significant differences were found for search time and accuracy of direction estimation. However, in AR, the Halo yielded significantly lower estimation errors, than Wedges [16].

Spatial auditory attention guidance by rendering the sound source spatially was found to be efficient [14, 31]. However, multi-modal usage with additional visual or haptic cues is recommended [1, 26]. Spatial auditory cues can compensate for minor flaws in visual cues [9]. However, objects associated with sounds can also provide gaze guidance, even if the sound is not spatial or comes from a different direction [40]. Spatial sound was found to be particularly useful in dual-task settings with visually demanding primary tasks [21]. For search tasks, users were found to roughly identify the target area with auditory cues, and subsequently search the smaller area visually to locate the target [28].

Diegetic attention guidance cues that are part of the surrounding context are often used in storytelling [39]. Aspects like environmental cues [40] and character interactions [48] can be used as guidance. However, diegetic cues were less effective than overt hints in auditory [6] and visual guidance [44].

Gaze redirection in training was investigated in multiple works. Drawing students' attention to something while they are learning by adding hints to visual stimuli helps them to learn [13, 38, 53]. However, this is also influenced by the appeal of the target area [42]. For short distractions, guidance might be harmful due to obtrusiveness [55]. Attention guidance was also used to model the gaze paths of experts for novices. A significant change in gaze patterns was found during training and shortly afterward with unknown stimuli [12, 25]. However, no long-term effects could be demonstrated [45]. Expert-based gaze guidance was also found to improve observation-based movement learning [13] and spatial performance in a mental rotation test [37].

3 Technical Methods

A Microsoft HoloLens 2nd generation (HoloLens) was used to display the attention guidance cues. The HoloLens supports visual and auditory AR content, offers on-board eye tracking, and is standalone as well as optical see-through. The combination of these features was preferred in the clinical training environment for safety and didactic reasons. Furthermore, the HoloLens is already established in medical training [33] as well as for attention guidance [16].

All cues were developed in the game engine Unity (Unity Software Inc., USA). Eye tracking was integrated with the Mixed Reality Toolkit v3.0¹. The gaze direction and intersections with colliders, e.g. for the target, were used.

3.1 Dual-task scenario

For this work, a medical examination scenario is abstracted into a simplified dual-task scenario. Both tasks are sub-tasks of the overall patient examination, which are taught as two simultaneous aspects as part of the training. The medical use case was defined with a physician working in teaching.

The primary task is a physical examination, requiring the continuous visual-haptic recognition and classification of symptoms. The patient's face is not considered part of this task and may be outside the field of vision. This may include an examination, like palpation or auscultation of the abdomen, or treatment of, e.g. a lesion on the leg. While the face may provide important information about the patient's condition, it is not necessarily relevant to the subtask. Within this work, the primary task was abstracted into a virtual placeholder task to provide context and distraction during the interviews. Four numbered buttons had to be pressed in sequence and were rearranged once finished (see Fig.2). The task required hand interaction and visual attention to perform, which is intended to provide a basic presentation of the visual-haptic medical examination task.

The secondary task is patient monitoring, specifically observation of the face. Practitioners have to look at the face and interpret



Figure 2: Setup for the qualitative evaluation with no cue visible. The head (right) was used as the target, and the numbered buttons (left) were used to simulate a haptic task with the head outside of the field of view.

the facial expressions, regularly. Thus, a model of a head was selected as the target (see Fig. 2). It was located approximately 70° to the right of the user, placing it just outside of the field of view of the HoloLens for our primary task. During the expert evaluation, users were instructed to regularly look at the face, but no further task, e.g. recognition of facial expressions, was added to not distract from the cues.

3.2 Attention Guidance Cues

Three visual and three auditory cues were implemented and can be viewed in Figure 1 and the supplementary video.

3.2.1 Visual Cues. Each visual cue communicated the target location and the time since the cue has been running, which is equivalent to the time since the target was focused. To determine the cue position, the target location was converted to pixel coordinates in screen space and the intersection of the display edge and the target position vector was determined. Due to the small field of view of the HoloLens, all visual cues were placed in the mid-peripheral visual field, which is beneficial for cues to be observable but not disruptive [23]. All visual cues had a runtime of 20s. The beginning was subtle for the first half to enable users to look at the target on their own. Intensities were then increased significantly over the 10s. With each glance at the target, the running time and cues were reset.

The cue *Flicker* was inspired by subtle and peripheral guidance cues using alternating luminance changes [4]. These approaches can reliably influence gaze behavior and also showed positive influences in training settings [3]. The *Flicker* was realized as a gray-scale image on a camera-overlay canvas of a white circle with fading edges. This was displayed once every second for half a second. Using a custom vertex and fragment shader, the inverted brightness value for each pixel was mapped to transparency to achieve an imperceptible transition between the image and the real environment. For intensification, the opacity was increased from an alpha value of 0.3 to 1 over 10s. Then, the diameter was increased from 1cm to 3cm.

The cue *Shadow* was inspired by blurring and fading effects in subtle and overt image modifications [25, 30]. A semi-transparent veil was used to obscure the environment with a narrowing circular grayscale gradient moving toward the target. *Shadow* was implemented as a vertex and fragment shader with Unity's ShaderLab. The shader was applied to a full-screen image in the camera overlay. Based on the target distance for each pixel, a grayscale distance map

¹<https://github.com/microsoft/MixedRealityToolkit-Unity>; Retrieved on 22nd of January, 2025.

was generated with a gradient from low/transparent values around a circular cut-out that centers on the target. Larger distances led to higher/opaque values up to the upper brightness limit at high distances from the target, resulting in a growing gray area obstructing the view on anything but the target region. For intensification, the cut-out radius was reduced and the upper brightness limit was increased up to 0.5, starting from 0.

The cue *Halo* is based on other 3D Halo visualizations [16, 17]. For the *Halo*, a horizontal circle was drawn around the target in screen space with a circle segment visible at the screen border. The shape was created using a vertex and fragment shader applied to a camera-overlay image. The diameter of the circle increased based on the angle and distance to the target. However, to maintain easily perceptible curvatures regardless of the distance to the target, the change in size is not linear. To compensate for this reduced perceptibility of the target distance, color was introduced using a temperature scale with red for small distances and blue for large distances [54]. The angle between the viewing and target direction is mapped from $[0^\circ, 180^\circ] \rightarrow [0, 1]$ for blue and $[0^\circ, 180^\circ] \rightarrow [1, 0]$ for red. The *Halo* is intensified by increasing opacity and size. The transparency is decreased continuously from fully transparent to completely opaque. After half the time, the width of the ring, starting from the base width of 0.05 in Unity units, begins to increase by the time factor from 0 to 0.15.

3.2.2 Auditory Cues. The auditory cues were designed to represent the range from artificial to natural sound. All sounds were implemented as spatial audios, originating in the target. Spatial audio was shown in previous work to be sufficient in attention guidance [40], even in small search areas with improving performance for farther targets [31].

The cues have a length between 0.5s and 3s and the spacing is randomized between 5s and 10s. Spatialization was integrated using the Microsoft Spatializer plugin².

The most artificial *Beep* cue was a beeping sound, based on non-contextual audio cues [41]. Like a danger signal, this is intended to attract attention and thus encourage the test person to look towards the target.

A more diegetic technique was the cue *Sigh*, which played a variety of breathing sounds to imitate patient noise. Five audio tracks from Pixabay³ were used and randomly selected. The breathing sounds were not played as a constant background sound to prevent users from increasingly ignoring the sound due to a habituation effect [56].

The cue *Talk* represented the most diegetic sound, with incorporated breathing sounds as well as short spoken sentences. The same files as for *Sigh* were used, but a third of the occurrences were randomly replaced with speech. The audio files for the sentences were generated with Narakeet⁴. All sentences are short small-talk statements, e.g. about the weather or the examination (see Appendix A).

4 Qualitative Evaluation

To obtain in-depth feedback from the perspective of various specialist areas concerning suitability as a teaching tool, a qualitative evaluation was carried out. Six experts reviewed the cues: three from the field of medical training (2f, 1m) and three in the area of human-computer interaction (HCI, 3m). The medical experts all had extensive experience in practical medical teaching. One of the experts has completed additional training in theoretical medical didactics. Senior researchers in the field of AR were interviewed as HCI experts. All had previous experience in the conception and development of medical training applications with AR and VR. Participants received no monetary compensation.

Three of the interviews were conducted remotely and three in person. However, all participants experienced the cues on a HoloLens locally. All interviews were conducted by two people, one who took notes and one who led the discussion. An audio recording was made in addition. First, the background and objective of the interview were explained and informed consent was obtained from all participants. The procedure was then described and it was explicitly pointed out that the assessment should be made in terms of teaching suitability, not in terms of attention attraction. This instruction was repeated several times during the interview. Participants were asked to comment and ask questions about the cues aloud. Then, the experts were asked to start the first cue. The order of the auditory/visual group and the sequence of the individual cues followed a Latin square layout. The cue observation was divided into three intervals, the length of which was determined by the experts. In the first interval, the buttons should be pressed to simulate focusing on the target area with no view of the head. In the second interval, the experts were free to experiment with the cue. In the third interval, the participants were asked to look at the abdominal task and consciously perceive a complete run of the cue. After completion of the observation, the following questions were asked: Do you have any general questions/comments?; Which advantages do you see with the cue in terms of teaching suitability?; Which disadvantages do you see with the cue in terms of teaching suitability?; What bothered you about the cue?; What did you like about the cue? Participants were then asked for any further comments or questions. This procedure was repeated for all cues. One run of the expert review lasted around one hour.

5 Results

The experts' statements were transcribed from audio recordings of the interviews. All comments were coded, summarized, and categorized into advantages and disadvantages. The results were then paraphrased and translated into English. Results for the visual cues are listed in Table 1 and for the auditory cues in Table 2.

In general, all experts agreed that the concepts helped in drawing attention to the head repeatedly. However, this required a familiarization phase, for example, due to the subtlety (*Flicker*) or lack of understanding (*Shadow*, *Halo*). More explicit attributes, such as movement, were mentioned to achieve a stronger guidance effect. However, they also expressed concern that this could create an over-reliance on the guidance. For the visual cues, it was criticized that the stimulus itself can be viewed through eye movements, without moving the head towards the patient's face. In some cases, this led

²<https://github.com/microsoft/spatialaudio-unity>; Retrieved on 21st of January, 2025.

³<https://pixabay.com/sound-effects/>; Retrieved on 21st of January, 2025

⁴<https://www.narakeet.com/>; Retrieved on 21st of January, 2025

Table 1: Summary of the expert review results for the visual cues. Feedback from the HCI experts was labeled H1-H3, responses from the medical didactics experts are marked C1-C3

Positive	Negative
<i>Flicker</i>	
• Natural looking and subtle stimulus (H1, H2) • Flashing supports perception and serves as a reminder (H2, H3, C2) • Precise directional guidance in all directions without occlusion problems due to small size (H1, H2, H3)	• Similar to perceptual issues or reflections and artifacts in the HoloLens (H1, H3, C1) • Perceptibility dependent on ambient light, hard to see in bright environments (H1, H2, H3, C2, C3) • Attracts little attention, easy to overlook or ignore (H1, H2, C1, C2, C3) • Demotivating effect, acts like an error indicator (H2)
<i>Shadow</i>	
• Easily perceptible in dark surroundings (H1, C3) • Precise directional instructions on where to look (H1) • Subtle, minimally distracting guidance in bright surroundings (H2, C2) • Gentle, non-punishing guidance (H2, H3, C1)	• Hard to see at the start, looks like a glitch (H2, H3, C1) • Hard to use without explanation (H2, H3, C1, C2, C3) • Can restrict the view of the primary task, which is disruptive and irritating when practicing (H1, C3) • Perceptibility dependent on ambient light (C1, C2)
<i>Halo</i>	
• Clear and comprehensible gaze guidance towards the face (H1, H2, H3, C2, C3) • Color attracts attention, has a comprehensible warning light effect, and is fun to look at (H1, H3, C1, C2) • Increasing intensification is good, easy to notice over time without being forced to look at the face (H2, C2) • Discreet at the start and not too distracting throughout (H1, H2) • Easy to see without occlusion problems (H1, H3, C2) • Does not interfere with the task or the view of the buttons (H3, C2, C3)	• Prompts playing and experimenting with the cue, can be distracting (H1, C1) • Looks like an artifact or reflection when the intensity is low (H1, C2, C3) • Initially difficult to understand because there are many parameters (H1) • The perception of colors can be difficult in some environments or with color perception disorders (H1, H2) • Could lead to over-reliance due to the overt design (H1) • The cue is cut off due to the small field of view of the HoloLens (H2, C1, C3)

to the cue being observed but having no guidance effect. For the auditory cues, the independence of the sounds from gaze behavior was criticized, which led to the stimuli being interpreted as error signals.

5.1 Discussion

Based on the feedback from the qualitative evaluation, recommendations for the individual cues are outlined below.

Flicker was not well visible in bright surroundings. While this fits for a subtle cue, it was noted by the experts that the flicker is very easy to ignore, particularly because of the similarity with display artifacts and reflections, which must also be disregarded. Further consideration in actual dual-task scenarios is necessary here. Different frequencies and transparency gradients can be considered to make the cue more noticeable but not annoying[47]. *Shadow* was very efficient for guidance, but the primary task could not be seen in dark environments. An ambient light-dependent adjustment of brightness and transparency would be useful here. *Halo* was described as well perceptible and not bothersome overall.

Movement towards the target would facilitate interpretation [18] but has proved confusing in related work [54]. Increased transparency could counteract a possible over-reliance on the cue [36]. The fade-in of the halo should be examined in more detail as the positive effect for low interference with primary tasks depends on cue design and use[24]. In general, future studies should consider the combination of design parameters of the visual cues and further investigate how components such as color or transparency affect attention guidance.

Beep was found unsuitable due to too many other beeping sounds in medical environments. Previous work has shown that very high alarm rates lead to a drop in the response rate [56]. Furthermore, user enjoyment was low. Other non-diegetic sounds, e.g. from musical instruments, could be investigated further here. The breathing sounds and sentences of *Sigh* and *Talk* were criticized. Constant diegetic patient sounds such as quiet breathing, throat clearing, or sighing were suggested instead. However, related work found diegetic sounds to be less helpful compared to non-diegetic cues [6]. Future studies should therefore consider the effect of creating a

Table 2: Summary of the expert review results for the auditory cues. Feedback from the HCI experts was labeled *H1-H3*, responses from the medical didactics experts are marked *C1-C3*

Positive	Negative
<i>Beep</i>	
• Simple, clear, and easy to understand (<i>H3, C2</i>) • Healthcare professionals are highly trained to respond to beeps (<i>H3, C1</i>) • Spatial sound is perceived well (<i>H1, C2</i>)	• Unpleasant sound, like an error-indicator (<i>H1, H2, H3, C1, C2, C3</i>) • No contextual connection to the face (<i>H1, H2, C3</i>) • Triggers a gaze movement, but does not fully guide to the face, too short, too infrequent (<i>H1, H2, C1</i>) • May interfere with other alarm sounds in a medical environment (<i>C2, C3</i>)
<i>Sigh</i>	
• Realistic, no artificial sound, thus more immersive (<i>H3, C1, C2, C3</i>) • Sustainable learning through better transferability (<i>C2</i>) • The nature of the sound triggers the gaze to the head and encodes urgency (<i>H2, H3, C2, C3</i>) • Volume and frequency are well (<i>C2</i>)	• May train response to pain sounds rather than general monitoring of the face (<i>H1, H2, C2, C3</i>) • Sound is quiet and monotonous, difficult to place spatially, and not a strong trigger (<i>C1</i>) • Sounds themselves are not realistic (<i>H2, C1, C2, C3</i>)
<i>Talk</i>	
• A realistic design of an AR patient could lead to immersive training (<i>H1, H3, C2, C3</i>) • Talking is close to the clinical situation (<i>C2, C3</i>) • Including breathing sounds is immersive (<i>H3, C1</i>)	• Breathing sounds are unnatural (<i>H1, H2, H3, C1, C3</i>) • Spatial sound is not well perceptible and too quiet (<i>H1</i>) • Sentences are unrealistic/ not representative of normal patient behavior (<i>H2, H3, C1, C2, C3</i>) • Distraction that is tuned out (<i>H2, C1, C3</i>) • May make the training obsolete if the problem does not exist with real patients (<i>H1</i>)

patient simulation, which could also help the primary task, on the learning success of attention allocation.

Only separate visual and auditory cues were included in the comparison. Related work found multimodal cues to be especially helpful in high load settings [7]. Furthermore, haptic guidance methods have proven to be effective in dual-task settings [5]. While this was excluded for this work to avoid additional hardware, this could be supplemented by other equipment. A comparison specifically for learning tasks would therefore be of interest in the future with various devices. Video see-through or VR devices should also be taken into account, which enable a better cue display. In particular, a small field of view, as with the HoloLens, was disadvantageous in related work [49].

The attentional tunneling effect of AR also applies specifically to attention guidance where targets that have not been cued are overlooked [57]. Using more subtle cues prevents interruptions and distractions [36] and could also lead to less reliance on the learning aids. When revising the cues, the cues should be made as discreet as possible.

While a dual-task setting was provided, the abstraction presented was not realistic. This is partly because an interview was conducted at the same time. We think that the qualitative results are useful for

further development, but also for an initial assessment of training applications. However, a more precise reproduction of the dual-task scenario is important in future quantitative user studies.

The actual users, e.g. medical students, were not included in the qualitative evaluation. Applicable results were ensured by experts with a medical-didactic background. For subsequent quantitative user studies, however, care must be taken to recruit from the target population. In addition, a more diverse group of didactic experts beyond medicine could further support the generalizability of the results in future evaluations.

6 Conclusion

Three visual and three auditory attention guidance cues were implemented for dual-task training scenarios in a medical use case. The cue *Halo* was preferred of the visual cues due to the clear guidance without much occlusion. However, there are concerns about the possible over-reliance of users as the cue is explicitly perceptible. The subtle cue *Flicker* was difficult to perceive depending on the environment. *Shadow* sometimes severely interfered with the primary task. A more realistic version of *Sigh* was devised for the auditory cues. *Talk* was found to be rather unrealistic, whereas *Beep* interfered too much with the clinical environment.

In general, most cues were found to be suitable for training attention allocation in dual task settings, requiring only minor design changes for improvement. In the next step, the tools will be revised and compared in a quantitative user study.

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A Phrases used for the cue *Talk*

- We're having a barbecue at the weekend.
- Tonight we're having stew.
- We had casserole for lunch.
- It's been raining all day today.
- Yesterday was much nicer weather.
- It's forecast to be really warm tomorrow.
- How much longer is this going to last?
- What are you doing?
- You have to press over here.
- Are you new here?
- Is your colleague on vacation?
- We want to go to Poland in the summer.
- Have you had a long day today?
- You look a bit tired.
- I've never seen you here before.